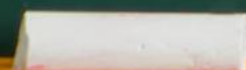


GRADE 11



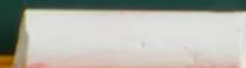
General Mathematics

Term 1 Week 4



Learning Competencies

- ✓ Differentiate sequences from series.
- ✓ Illustrate arithmetic and geometric series.
- ✓ Represent series using sigma notation and vice versa
- ✓ Find the sum of arithmetic and geometric series.
- ✓ Solve problems involving arithmetic and geometric series.

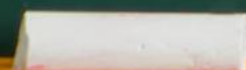


Activity B.2:



Lesson Discussion:

SERIES



Key Terms and Definitions

SERIES

The sum of the terms of a sequence.

Example: For the sequence 1, 3, 5, the series is $1 + 3 + 5 = 9$

FINITE SERIES

A series with a definite number of terms.

Example: $2 + 4 + 6 + 8$ (only 4 terms)

INFINITE SERIES

A series that continues without end.

Example: $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots$

The n th term of a sequence is usually denoted by a_n



Key Terms and Definitions

PARTIAL SUM (S_n)

The sum of the terms of a sequence.

If the n th term of the sequences is a_n , then $S_n = a_1 + a_2 + a_3 + \dots + a_n$

Example: In the sequence, 1, 3, 5, 7, 9, . . . ,

$$S_4 = 1 + 3 + 5 + 7 = 16$$

S_n is the n th term of the series

SIGMA NOTATION (Σ)

A shorthand symbol used to express a series.

To write $a_1 + a_2 + a_3 + \dots + a_n$ in sigma notation:

$$\sum_{n=1}^n a_n$$



In general,

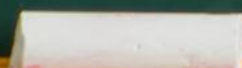
$$\sum_{n=start}^{k=end} a_n$$

(expression in terms of a_n)

Example: $1 + 2 + 3 + 4 + 5$ in sigma notation is

$$\sum_{n=1}^5 a_n$$



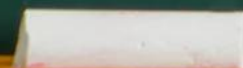


Activity B.3:



Practice

Problems- Sum It



Activity B.3: Practice Problems- Sum It

Instructions: Do the following.

1. Find the sum of the first 4 terms of the sequence,
3, 7, 11, 15, . . .

Answer: $3 + 7 + 11 + 15 = 36$

2. Find the sum of the sequence:
5, 10, 20, 40, 80

Answer: $5 + 10 + 20 + 40 + 80 = 155$



Activity B.3: Practice Problems- Sum It

3. Write the series, $2 + 4 + 6 + 8 + 10$ in sigma notation.

Answer: The terms are 2×1 , 2×2 , 2×3 , 2×4 , 2×5 .

There are 5 terms in the series.

In sigma notation:

$$\sum_{n=1}^5 2n = 30$$



Activity B.3: Practice Problems- Sum It

4. Evaluate the series written in sigma notation:

$$\sum_{n=1}^4 (3n + 1)$$

Answer: When $n = 1$, $3(1) + 1 = 3 + 1 = 4$

When $n = 2$, $3(2) + 1 = 6 + 1 = 7$

When $n = 3$: $3(3) + 1 = 9 + 1 = 10$

When $n = 4$: $3(4) + 1 = 12 + 1 = 13$

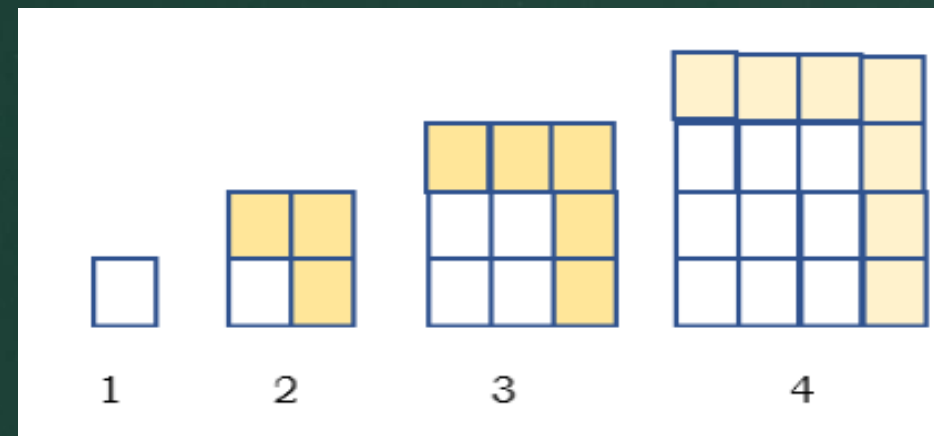
Then add: **$4 + 7 + 10 + 13 = 34$**

$$\sum_{n=1}^4 (3n + 1) = 34$$



Activity B.3: Practice Problems- Sum It

5. The figure shows a growing pattern of tiles. Count the number of tiles in each figure.



- How many tiles are there in the 5th figure?
- How many tiles are there in total from the 1st to the 7th figures?
- What is the rule used to determine the number of tiles in each figure?

Answer:

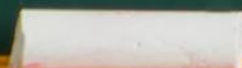
- 25 tiles
- 140 tiles in total
- There are n^2 tiles in Figure n





Activity C.1:

Saving for a Goal



Activity C.1: Saving for a Goal

Scenario:

Anna decides to save money every week to buy a bicycle worth ₱3,000. She starts by saving ₱100 in the first week and increases her savings by ₱50 each week.

Questions:

1. How much will Anna save in the 5th week?
2. What is the total amount saved after 5 weeks?
3. Will she reach her goal in 5 weeks? If not, how many weeks will it take?
4. How much will Anna save if she saves for 20 weeks? (You may use an Excel spreadsheet in finding the sum)



Activity C.1: Saving for a Goal

ANSWER:

Show the calculations step by step and adding terms.

Let a_n be the amount saved in week n .

$$a_1 = 100$$

$$a_2 = 150$$

$$a_3 = 200$$

$$a_4 = 250$$

$$a_5 = 300$$

Anna's savings per week is:

$$\text{1st week} = \text{P}100$$

$$\text{2nd week} = \text{P}100 + \text{P}50 = \text{P}150$$

$$\text{3rd week} = \text{P}150 + \text{P}50 = \text{P}200$$

$$\text{4th week} = \text{P}200 + \text{P}50 = \text{P}250$$

$$\text{5th week} = \text{P}250 + \text{P}50 = \text{P}300$$



Activity C.1: Saving for a Goal

ANSWER:

Add her savings:

Php 100 + Php 150 + Php 200 + Php 250 + Php 300 = Php 1000

Total amount saved is Php 1000

$$S_n = a_1 + a_2 + a_3 + a_4 + a_5$$

In the 5th week she will save Php 300

Anna will save Php 1000 in 5 weeks

No, she will not reach her goal in only 5 weeks. Anna will need 10 weeks to reach Php 3000.

Php 11,500 if she saves for 20 weeks.

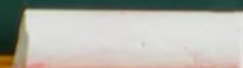




Activity B.2:

Lesson Discussion:

Arithmetic Sequence and Series

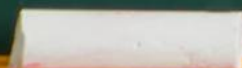


Key Terms and Definitions

Sequence – A set of numbers arranged in a specific order or pattern.

Series – The sum of the terms in sequence.

Term – It refers to each of the elements in a sequence or series.



Key Terms and Definitions

Arithmetic Sequence

It is a special type of number pattern in which every term after the first is obtained by adding a number to the preceding term.

This means that in an arithmetic sequence the difference between two consecutive terms is constant.

First term (a_1) – the very first value in the sequence. It is the starting point from which other terms in the sequence are produced.

n th term – it refers to the general or specific term at the position n in the sequence.

Common Difference – it is the constant difference between the two consecutive terms in an arithmetic sequence. It is also the number added to a term to get the next term.

Arithmetic Series – it is the sum of terms of an arithmetic sequence. It is denoted by S_n .

S_n is the sum of the first n th terms of a sequence.

$$S_n = a_1 + a_2 + \dots + a_n$$



Arithmetic Sequence

Arithmetic Sequence is a special type of number pattern in which the two consecutive terms have a constant difference called the common difference (d).

To Calculate the nth term of an arithmetic sequence, use the formula:

$$a_n = a_1 + (n-1)d$$

where:

a_n = the nth term

a_1 = the first term of the arithmetic sequence

d = the common difference

n = the position of the term in the sequence

Arithmetic Sequence

Example 1:

Find the 10th term of the arithmetic sequence: 5, 8, 11, 14, ...

Solution:

Step 1: Identify the value of the variables of the formula.

$$a_1 = 5$$

$$d = 3$$

$$n = 10$$

$$a_n = ?$$

Step 2: Substitute the values to the formula. Then, solve for a_n .

$$a_{10} = 5 + (10 - 1)3$$

$$= 5 + (9)3$$

$$= 5 + 27$$

$$a_{10} = 32$$

Therefore, the 10th term in the arithmetic sequence is 32.

Arithmetic Sequence

Example 2:

In the arithmetic sequence: 3, 9, 15, 21, ... ; find the value of n if $a_n = 75$.

Solution:

Step 1: Identify the value of the variables of the formula.

$$a_1 = 3 \qquad d = 6$$

$$a_n = 75 \qquad n = ?$$

Step 2: Substitute the values to the formula. Then, solve for n .

$$a_n = a_1 + (n-1)d$$

$$75 = 3 + (n-1)6$$

$$75 = 3 + 6n - 6$$

$$75 - 3 + 6 = 6n$$

$$78 = 6n$$

$$n = 13$$

Therefore, 75 is in the 13th term of the arithmetic sequence.

Arithmetic Sequence

Example 3:

The 5th term of an arithmetic sequence is 20, and the 12th term is 48. Find the following: (a) common difference, (b) first term, and (c) the 30th term.

Solution:

Step 1: Identify the given values.

$$a_5 = 20$$

$$a_{12} = 48$$

Step 2: Substitute the values to the formula.

$$a_n = a_1 + (n-1)d$$

$$\text{For } a_5 = 20 : 20 = a_1 + (5 - 1) d$$

$$20 = a_1 + 4d \text{ (Equation 1)}$$

$$\text{For } a_{12} = 48 : 48 = a_1 + (12 - 1) d$$

$$48 = a_1 + 11d \text{ (Equation 2)}$$

Step 3: Subtract the equations

$$20 = a_1 + 4d$$

$$\underline{48 = a_1 + 11d}$$

$$-28 = -7d$$



$$d = 4$$

Arithmetic Sequence

Solution:

Step 4: Substitute the value of $d = 4$ to equation 1:

$$20 = a_1 + 4(4)$$

$$20 = a_1 + 16$$

$$20 - 16 = a_1$$

$$\mathbf{a_1 = 4}$$

Step 5: Solve for the 30th term.

$$a_{30} = 4 + (30 - 1) 4$$

$$= 4 + (29)4$$

$$= 4 + 116$$

$$\mathbf{a_{30} = 120}$$

Therefore, the common difference is 4, first term is 4, and the 30th term is 120

Arithmetic Sequence

Example 4:

During the dry season, a farmer in your town is testing a new water irrigation system for his rice field. After 1 minute of turning on the water pump, the water level in the irrigation canal rose to 30 centimeters. After 2 minutes, it reached 85 centimeters, and after 3 minutes, the level was 140 centimeters. If the water level continues to rise at this constant rate, how high will the water level be after 16 minutes?

Solution: (Let a_n = the water level after n minutes.)

Step 1: Identify the given

$$a_1 = 30\text{cm}$$

$$n = 16\text{min}$$

$$d = 55\text{cm}$$

$$a_n = ?$$

Step 2: Substitute the values to the formula. Then, solve for a_n .

$$a_{16} = 30 + (16 - 1)55$$

$$= 30 + (15) 55$$

$$= 30 + 825$$

$$a_{16} = \mathbf{855 \text{ cm}}$$

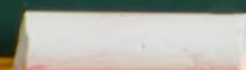
Therefore, after 16 minutes, the water level will be 855 centimeters high in the irrigation canal.

Arithmetic Series

Arithmetic Series is the sum of terms of an arithmetic sequence. It represents adding the terms of the sequence together.

Let's say we have the arithmetic sequence: 3, 6, 9, 12. This is an arithmetic sequence where the common difference $d = 3$. Now, let's add the terms: $3 + 6 + 9 + 12 = 30$. The sum is an example of arithmetic series.

Let us compute the sum of the first n terms of an arithmetic series.



Arithmetic Series

To Calculate the n th term of an arithmetic series, use the formula:

$$S_n = a_1 + a_2 + \dots + a_n(n-1)d$$

$$S_n = a_1 + (a_1 + d) + (a_1 + 2d) + \dots + [a_1 + (n-1)d]$$

There are n terms, so we add together all the a_1 's

$$S_n = na_1 + [1 + 2 + \dots + (n-1)]d$$

We have already derived the formula for the sum of the numbers from 1 to $n-1$.

$$S_n = na_1 + (n-1)nd/2 = n[2a_1 + (n-1)d]/2$$

Now we know that $a_1 + (n-1)d = a_n$. Therefore,

$$S_n = n(a_1 + a_n)/2$$

Arithmetic Series

We therefore have the following formulas to compute the sum of the first n terms of an arithmetic series:

If the last n th term (a_n) is given:

$$S_n = \frac{n}{2}(a_1 + a_n) \quad \text{or}$$

If the last n th term (a_n) is not given:

$$S_n = \frac{n}{2}[2a_1 + (n - 1)d]$$

where:

S_n = the sum of first n terms

a_1 = first term

a_n = last n th term

d = common difference



Arithmetic Series

Example 1:

Find the sum of the arithmetic sequence with 10 terms: 3, 10, 17, ..., 66.

Solution:

Since the last n th term is given, use the **1st formula**. $S_n = \frac{n}{2}(a_1 + a_n)$

Step 1: Identify the value of the variables of the formula.

$$a_1 = 3$$

$$a_n = 66$$

$$n = 10$$

$$S_n = ?$$

Step 2: Substitute the values to the formula.

$$S_{10} = \frac{10}{2}(3 + 66)$$

$$= 5(69)$$

$$\mathbf{S_{10} = 345}$$

Therefore, the sum of the arithmetic sequence is 345.

Arithmetic Series

Example 2:

Find the sum of the first 15 terms of the arithmetic sequence: 4, 9, 14, 19, ...

Solution:

Since the last nth term is not given, use the **2nd formula**. $S_n = \frac{n}{2} [2a_1 + (n - 1)d]$

Step 1: Identify the value of the variables of the formula.

$$a_1 = 4$$

$$d = 5$$

$$n = 15$$

$$S_n = ?$$

Step 2: Substitute the values to the formula.

$$\begin{aligned} S_{12} &= \frac{15}{2} [2(4) + (15 - 1)5] \\ &= 7.5 [8 + (14)5] \\ &= 7.5 [8 + 70] \\ &= 7.5 [78] \end{aligned}$$

$$\mathbf{S_{12} = 585}$$

Therefore, the sum of the arithmetic sequence is 585.

Arithmetic Series

Example 3:

Mang Tonyo is a kalesa driver in the historic streets of Vigan, Ilocos Sur. On his first day, he earns ₱200. As more tourists visit throughout the week, his earnings increase by ₱100 each day. If this pattern continues for 15 days, how much will Mang Tonyo earn in total?

Solution: (Let a_n = the earning of Mang Tonyo on day n)

$$S_n = \frac{n}{2} [2a_1 + (n - 1)d]$$

Since the last earning or n th term is not given, use the **2nd formula**.

Step 1: Identify the given values.

$$a_1 = \text{₱}200$$

$$d = \text{₱}100$$

$$n = 15 \text{ days}$$

$$S_n = ?$$

Step 2: Substitute the values to the formula.

$$\begin{aligned} S_{15} &= \frac{15}{2} [2(200) + (15 - 1)100] \\ &= 7.5 [400 + (14)100] \\ &= 7.5 [400 + 1400] \\ &= 7.5 [1800] \end{aligned}$$

$$\mathbf{S_{15} = \text{₱}13,500}$$

Therefore, Mang Tonyo will earn a total of ₱13,500 in 15 days.

Activity B.2: Understanding the Geometric Sequence and Series

GEOMETRIC SEQUENCE is a special type of sequence where each term after the first is obtained by multiplying the preceding term by a fixed non-zero number.

The fixed multiplier is called the common ratio, denoted by r . It is called the common ratio because it is the ratio of any two consecutive terms in the sequence.

$$r = a_{n+1}/a_n$$

The formula in calculating the n th term in a geometric sequence is

where:

a_n = the last n th term

a_1 = first term

r = common ratio

n = the position of the term

$$a_n = a_1 r^{n-1}$$



Activity B.2: Understanding the Geometric Sequence and Series

To calculate the n th term of a geometric sequence, let's follow the steps:

1. Identify the value of the first term.
2. Find the value of the common ratio.
3. Apply the formula in calculating the n th term.

$$a_n = a_1 r^{n-1}$$



Geometric Sequence

Example 1:

Find the 5th term of a geometric sequence where the first term is 5 and the common ratio is 3.

Solution:

$$a_1 = 5$$

$$n = 5$$

$$r = 3$$

$$a_n = ?$$

$$a_n = a_1 r^{n-1}$$

$$a_5 = 5(3)^{5-1}$$

$$= 5(3)^4$$

$$= 5(81)$$

$$\mathbf{a_5 = 405}$$

Therefore, 405 is the value of the 5th term in the sequence.

Geometric Sequence

Example 2:

Find the 10th term of the geometric sequence: 4, 12, 36, ...

Solution:

$$a_1 = 4$$

$$n = 10$$

$$r = 3$$

$$a_n = ?$$

$$a_n = a_1 r^{n-1}$$

$$a_{10} = 4(3)^{10-1}$$

$$= 4(3)^9$$

$$= 4(19683)$$

$$a_{10} = 78,732$$

Geometric Sequence

Example 3:

The 4th term of a geometric sequence is 64, and the 9th term is 486. Find (a) the common ratio, (b) the first term, and (c) the 7th term.

Solution:

From a_4 :

$$64 = a_1 r^{4-1}$$

$$64 = a_1 r^3 \text{ (Equation 1)}$$

From a_9 :

$$486 = a_1 r^{9-1}$$

$$486 = a_1 r^8 \text{ (Equation 2)}$$

Divide the equation 2 by the equation 1:

$$486/64 = a_1 r^8 / a_1 r^3$$

Solving for the common ratio (r).

$$243/32 = r^5$$

$$\mathbf{3/2 = r}$$

Substitute $r=3/2$ in equation 1:
Solving for the first term (a_1)

$$64 = a_1 \left(\frac{3}{2}\right)^3$$

$$\frac{512}{27} = a_1$$

Geometric Sequence

Example 3:

The 4th term of a geometric sequence is 64, and the 9th term is 486. Find (a) the common ratio, (b) the first term, and (c) the 7th term.

Solution:

$$\begin{aligned} a_4 &= 64 & r &= 3/2 \\ n &= 9 & a_n &= ? \end{aligned}$$

$$a_n = a_1 r^{n-1}$$

$$a_7 = \frac{64}{81} \left(\frac{3}{2}\right)^{7-1}$$

$$= \frac{64}{81} \left(\frac{3}{2}\right)^6$$

$$= 216$$

Geometric Sequence

Example 4:

A local farmer named Mang Pedro started raising native chickens. In his first month, he had 3 chickens. Every month, the number of chickens tripled as each chicken hatched more chicks. The number of chickens each month follows a geometric sequence.

Problem:

1. How many chickens does Mang Pedro have by the 5th month?
2. How many chickens does he have by the 8th month?
3. Write the general formula to find the number of chickens in the n th month.

Geometric Sequence

Solution:

Given: First term $a_1=3$ Common ratio $r = 3$

1. 5th month

$$a_5 = 3 (3)^{5-1} = 3 (3)^4 = 3 (81)$$

$$a_5 = \mathbf{243 \text{ chickens}}$$

2. 8th month

$$a_8 = 3 (3)^{8-1} = 3 (3)^7 = 3 (2187)$$

$$a_8 = \mathbf{6561 \text{ chickens}}$$

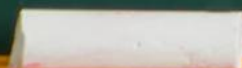
3. General Formula

$$a_n = 3 \cdot (3)^{n-1}$$

$$a_n = 3^n$$



Geometric Series



Geometric Series

Review of terms:

1. First Term (a_1): This is the very first term of the sequence. It's the starting value from which all other terms are derived by multiplication with the common ratio.
2. Common Ratio (r): This is the constant factor by which each term in the sequence is multiplied to get the next term. It is the ratio of all consecutive terms in the series.
4. S_n : This is the sum of the first n terms of a geometric sequence.



Geometric Series

To derive the formula for the sum of the first n terms of a geometric sequence.

A geometric sequence has:

First term: a_1

Common ratio: r

Number of terms: n

When $r \neq 1$, let:

$$S_n = a_1 + a_1r + a_1r^2 + a_1r^3 + \dots + a_1r^{n-1} \quad \text{eq. (1)}$$

Now, multiply both sides of equation (1) by r :

$$rS_n = a_1r + a_1r^2 + a_1r^3 + \dots + a_1r^n \quad \text{eq. (2)}$$

Next, subtract equation (2) from equation (1):

$$\begin{array}{r} S_n = a_1 + a_1r + a_1r^2 + a_1r^3 + \dots + a_1r^{n-1} \\ - \\ rS_n = a_1r + a_1r^2 + a_1r^3 + \dots + a_1r^n \\ \hline \end{array}$$

$$S_n - rS_n = a_1 - a_1r^n$$

Extract S_n as a common factor on the left-hand side of the equation and a_1 on the right-hand side.

$$S_n(1 - r) = a_1(1 - r^n)$$



Geometric Series

Extract S_n as a common factor on the left-hand side of the equation and a_1 on the right-hand side.

$$S_n(1 - r) = a_1(1 - r^n)$$

Now, divide both sides $1 - r$.

$$S_n = \frac{a_1(1 - r^n)}{(1 - r)}$$

Therefore, if $-1 < r < 1$, the sum of an infinite geometric series is

$$S = \frac{a_1}{1 - r}$$



Geometric Series

Example 1:

Find the sum of the first five terms of the geometric series: $2 + 6 + 18 + 54 + \dots$

Solution:

$$a_1 = 2$$

$$n = 5$$

$$r = 3$$

$$S_n = ?$$

$$S_5 = 2 \left(\frac{1 - 3^5}{1 - 3} \right)$$

$$S_5 = 2 \left(\frac{1 - 243}{1 - 3} \right)$$

$$S_5 = 2 \left(\frac{-242}{-2} \right)$$

$$S_5 = 2 (121)$$

$$S_5 = 242$$

Geometric Series

Example 2:

Consider the geometric series: $10 + 5 + 2.5 + 1.25 + \dots$ Find the sum of the infinite series.

Solution:

$$a_1 = 10$$

$$r = 1/2$$

$$S_n = ?$$

$$S_n = \frac{10}{1 - \frac{1}{2}}$$

$$S_n = \frac{10}{\frac{1}{2}}$$
$$S_n = 10 \times 2$$

$$S_n = 20$$

Geometric Series

Example 3:

Mang Tonyo owns a small sari-sari store in the barangay. One day, he decided to start saving coins in an alkansya (piggy bank). On the first day, he puts ₱20. Each following day, he doubles the amount he saves from the previous day. He plans to continue this for 7 days. How much total money will Mang Tonyo have saved in the alkansya after 7 days?

Solution: Let a_n = The money Mang Tonyo will save in the alkansya on the n th day.

$$a_1 = 20$$

$$r = 2$$

$$n = 7$$

Geometric Series

Example 3:

Use the Geometric Series Formula:

$$S_7 = a_1 \frac{1 - r^n}{1 - r}$$

$$S_7 = 20 \left(\frac{1 - 2^7}{1 - 2} \right)$$

$$S_7 = 20 \left(\frac{1 - 128}{1 - 2} \right)$$

$$S_7 = 20 \left(\frac{-127}{-1} \right)$$

$$S_7 = 20(127)$$

$$S_7 = \text{₱}2540$$

Therefore, Mang Tonyo will save ₱2540 in 7 days.

Thank You

